

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2024

Subject: 031001111 Mechanics I

Section: 15-18

Date: 30 September 2024

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 6 pages (including this page) and they are worth 35 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

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Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Part I	Part II
CLO 5	Able to draw diagrams of independent objects and explains Newton's laws of motion. Applying Newton's laws of motion to objects moving horizontally and vertically and finding related quantities.	1-3	1-3
CLO 6	Analyze and calculate the force of friction between the touching surfaces of a pair of objects and find the coefficient of friction between the contact surfaces.		4,8
CLO 7	Able to consider the net force and explain the balance of three forces.	4-5	5-6
CLO 8	Describe the mechanical balance of an object moments and the sum of moments that affect sliding, rotation, coupling forces, and the effect of coupling forces on the balance of objects.		7

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Part I Multiple choices.

Please choose the correct answer for each question. (5 points)

1. If an object of mass m moves with constant speed. What is net force of the object?
 a. 0 b. ma c. mg d. constant

2. Two forces act on an object given by $F_1 = 60 \text{ N}$ and $F_2 = 80 \text{ N}$ where the two forces are perpendicular to each other. Find net force of the object?
 a. 20 N b. 90 N c. 100 N d. 140 N

3. Identify the incorrect statement about Action and Reaction?
 a. they act on different bodies.
 b. action is cause and reaction is effect.
 c. has equal magnitude and opposite direction.
 d. action and reaction are forces which acts at same instant.

4. If an object is in equilibrium, which of the following statement is not true?
 a. The speed is constant. b. The net force is zero.
 c. The net force is equal ma . d. The acceleration is zero.

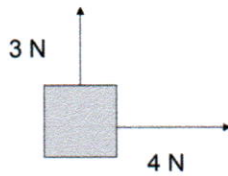
5. A light string passing over a smooth light pulley connects two blocks of masses m and M vertically. If the acceleration of the system is $g/8$, then the ratio of masses is
 a. 8/1 b. 9/7 c. 4/3 d. 5/3

Part II Writing

1. Planet A has a gravitational force 2 times that on the earth. How much would weight of a 60 kg man on the planet A? (2 points)

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2. From the figure, find net force? (2 points)



3. A 10 kg box moving with an acceleration 2 m/s^2 . Find

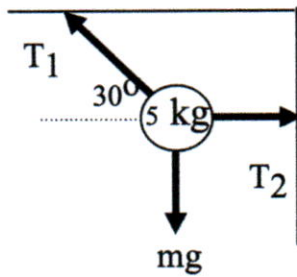
(a) Magnitude of net force acting on it (2 points)

(b) If you place a box weighing 10 kg on top of it and use the same force to push it find the acceleration of the object. (2 points)

4. A bullet of mass $1 \times 10^{-2} \text{ kg}$ moving with a speed of 80 m/s enters a wooden block and is stopped after 0.4 m . What is the average resistive force exerted by the block on the bullet? (4 points)

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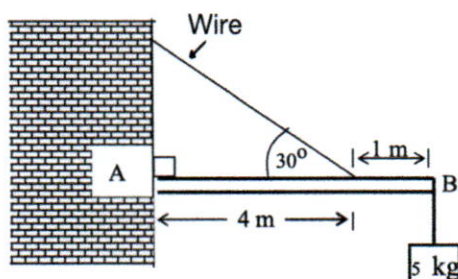
5. From the figure, if the system is in equilibrium, find the magnitude of the tension in the rope T_1 and T_2 ? (4 points)



6. A 60 kg adult sits at one end of 10 m long board. His friend 42 kg sits on the other end. Where should the pivot be placed so that the board is balanced? (Ignoring the board's mass) (4 points)

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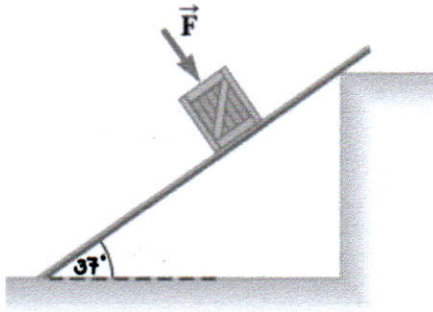
7. A wire is pulling a beam AB with a mass of 5 kg and is suspended at end B. If a uniform beam has a weight 20 N and a length of 5 m and end A is fixed to a wall, the beam is in equilibrium as shown in the figure. Find the tension in the wire in Newtons. (4 points)



8. A box 5 kg is placed on a 37-degree inclined plane. When a force F is applied to it, the box stops moving and does not sliding down. If the coefficient of static friction between a box and the incline surface is 0.3. Find

a. Normal force (N) (3 points)

b. Force (F) (3 points)



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